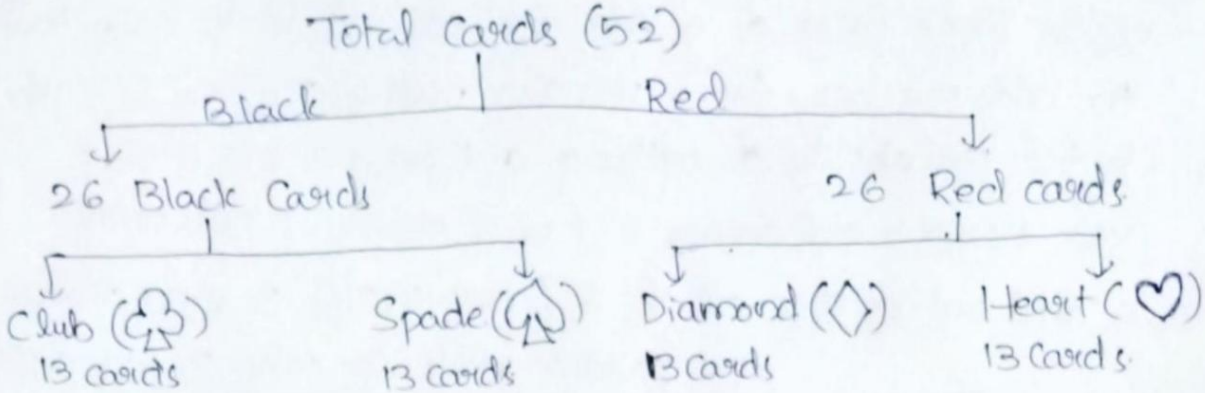


Conclusion:- (i) 13 Black club cards (A,2,3,4,5,6,7,8,9,10, J,Q,K)

(ii) 13 Black Spade cards (A,2,3,4,5,6,7,8,9,10, J,Q,K)

(iii) 13 Red Diamond cards (A,2,3,4,5,6,7,8,9,10, J,Q,K)

(iv) 13 Red Heart cards (A,2,3,4,5,6,7,8,9,10, J,Q,K).

Example 1:- The probability of choosing a heart from a deck of cards is given by

$$P(\text{heart}) = \frac{\text{No. of hearts}}{\text{total no. of cards}} = \frac{13}{52} = \frac{1}{4} = 0.25 \text{ [unlikely]}$$

Example 2:- The probability of choosing a three from a deck of cards is

$$P(\text{three}) = \frac{\text{No. of threes}}{\text{Total no. of cards}} = \frac{4}{52} = \frac{1}{13} = 0.077 \text{ [very unlikely]}$$

Example 3:- The probability of a two coming up after rolling a die (angular for dice) is

$$P(\text{two}) = \frac{\text{number of twos}}{\text{total no. of sides}} = \frac{1}{6} = 0.167 \text{ [very unlikely]}$$

Example 4:- A die is cast once what is the probability of getting either 6 or 3?

Ans:- 3 or 6 nothing but 2 possibilities out of total 6 possibilities.

$$P(3 \text{ or } 6) = \frac{3/6 = 2}{1/2/3/4/5/6 = 6} = \frac{2}{6} = \frac{1}{3} \text{ //. [0.33 = unlikely]}$$

Ex:5:- Three faces of a die are painted white such that only the odd number faces remain visible. The die is cast. What is the probability of getting a 1 (or) a 3 (or) a 5 ?

Ans:- possible out comes = 1 or 3 or 5 = 3 out comes

Total out comes = out of 6 (1|2|3|4|5|6) 3 even digited faces are painted white. So, only 3 odd digits we can see any one of them when die thrown. = 3

$P(1/3/5 \text{ when 3 even digited faces are white painted}) = \frac{3}{3} = 1$

[1 = very likely = certain]

Ex:6:- What is the probability that a dice with 12 faces relising a value = 9, ie each face has a number starting from 0.

Ans:- here dice has 12 faces & shows 0, 1, 2, 3, ... 11. So. the Probability of getting 9 is => Occurance of 9 is once

$$P(9) = \frac{1}{12} = \frac{\text{occurance of 9 only once}}{\text{Total possible out comes}}$$

Ex:7:- A couple decides to start a family & adopt a kid, After a year, they adopt another kid. what is the probability that the first kid is a girl?

Ans:- They may adopt any gender (either girl/boy) nothing but two possibilities.

But required one is occurance of /adopting a girl only one possibility.

So, $P(\text{adopting the first as girl}) = \frac{1}{2} = \frac{\text{Required}}{\text{Total.}}$

Ex:8:- In a family of 3 childrens, what is the probability that all the childrens in a family will be